Multi-Sensor Mental Health Detection

Final Year Project Report

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Project Details

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Area of specialization				
Project Group Members				
Sr.#	Reg. #	Student Name	Email ID	Signature
(i)	Group Leader	M. Saif	2022-bs-cs-233@tuf.edu.pk	
(ii)		Fakhir Baig	2022-bs-cs-195@tuf.edu.pk	
(iii)		Haseeb Ahmed	2022-bs-cs-185@tuf.edu.pk	

*The candidates confirm that the work submitted is their own and appropriate credit has been given where reference has been made to the work of others.

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This is to certify that, I am ________________________ S/D/O _______________________, group leader of FYP under registration no _________________ at the Department of Computer Science, The University of Faisalabad. I declare that my FYP proposal is checked by my supervisor and the similarity index is ________% that is less than 20%, an acceptable limit by HEC. The report is attached herewith as Appendix A.

Name of Group Leader: __________________________ Signature: ________________

Member Name: _________________________________ Signature: ________________

Member Name: _________________________________ Signature: ________________

Name of Supervisor: ____________________ Co-Supervisor (if any): _________________

Designation: ___________________________ Designation: _________________________

Signature: _____________________________ Signature: ___________________________

Program Head / Coordinator: ________________________________

Signature: ____________________________

Declaration

We here by declare that the content of this project report title Multi-sensor Mental Health Detection submitted to the department of computer science, is a documentation of a unique work we created under the supervision of Supervisor Ma'am Muqaddasa Jabeen and that no part has been plagiarized (except the references, some standard mathematical or genetic models/questions/protocols). Additionally, this project is presented in partial completion of the degree requirements for a Bachelor of Science in Computer Science. The University may take action if the above statement is found inaccurate at any stage.

Student Name	Registration No	Signature
Muhammad Saif	2022-bs-cs-233	
Fakhir Baig	2022-bs-cs-195	
Haseeb Ahmad	2022-bs-cs-185	

Supervisor: Ma'am Muqadasa Jabeen **Signature/Date**

Certificate

We accept the work contained in the report titled Multisensor Mental Health Detection, written by Muhammad Saif, Fakhir Baig, Haseeb Ahmad as a confirmation to the required standard for the partial fulfilment of the degree of Bachelor of Science in Computer Science

Approved by:

Supervisor: Ma'am Muqadasa

Signature: ____

Internal Examiner: Mr. Umer Latif

Signature: ____

External Examiner:

Signature: ____

FYP Coordinator: Dr. Uzair Saeed

Signature: ____

Program Head / Coordinator: Mr. Umer Lateef

Signature: ____

Dean FIT: Prof. Dr. Majid Hussain

Signature: ____

Date:

Abstract

Mental wellness is a crucial domain of human health. Accurate monitoring these conditions has historically been difficult since normal professional methods are too expensive. While dedicated EEG installations are constrained by their obtrusive nature, wearable sensors typically evaluate simple proxies for physical activity, making every day mental health tracking difficult. The goal of this project is to create an intelligent, multi-sensor system for the objective identification of mental health conditions such as stress, exhaustion, and cognitive load. The suggested approach combines cardiovascular responses (photoplethysmography, or PPG) and sympathetic arousal data (wrist-level electrodermal activity, or EDA) with EEG data to monitor the central nervous system. We overcome the limitations of single-sensor approaches by concurrently collecting and cooperatively processing multi-modal data using artificial intelligence and cutting-edge machine learning techniques. This system adds a self-validating framework to a real-time emotional wellness assessment and allows proactive mental health monitoring on a remote, continuous basis. We provide a self-validating system, which allows for dynamic assessment on emotional wellness in a more cost-effective manner and places user comfort at the forefront.

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List of Abbreviations

EDA	Electrodermal Activity
EEG	Electroencephalography
CNN	Convolutional neural network
DL	Deep learning
ML	Machine learning
FR	Functional Requirement
U-Net	Universal Network
GSR	Galvanic Skin Response
PPG	Photoplethysmography

CHAPTER 1: INTRODUCTION

1.1 Introduction

A critical aspect of human health, mental wellness is rarely assessed because both are expensive and difficult to monitor. While EEG and other professionally owned devices are either too costly or too complex for a person to use effectively on a daily basis, wearable, which are seen to be the most sensible solutions, just assess anaerobic and aerobic exercise, steps, or heart rate. The current project seeks to create an intelligent, multi-sensor system that is dependable and easily accessible for the identification of mental health conditions[1]. By using EEG as the main source of signals, in conjunction with wrist-level EDA and PPG data, the system can provide the user a comprehensive understanding of their levels of stress, fatigue and emotional balance.

1.2 Aim & Objectives

To investigate an advanced multi-sensor system for precise and instantaneous identification of both a user's stress and degree of mental fatigue using EEG, EDA, and PPG data.

1. To simultaneously record and process EEG, EDA, and PPG signals.
2. To implement machine learning models to classify caused stress and emotional states in real-time [2].
3. To create an interface for easy visualization of results.
4. To provide secure and private data transport, and storage.
5. To evaluate accuracy and usability of system.

1.3 Problem Statement

Mental Health Monitoring still proves to be a challenge because there are no inexpensive as well as precise real-time tools for the assessment of emotions. Most of the available tools for mental health monitoring are focused solely on one parameter of physiological signals, thus making it difficult for them to yield accurate results regarding stress[3]. Even advanced tools do exist in the market; however, they are quite costly and impractical for common usage by everyone.

Stress needs to be analyzed in a multi-sensor environment in order for the analysis to be accurate. Present solutions do not give a complete picture of emotional conditions in real- world environments. This makes early stress recognition and awareness about mental health less efficient[4]. The system will help to bridge this problem as it will be using the signals of EEG, EDA, as well as the signal from the heartbeat. This will increase the chances of accurate stress recognition.The system will be inexpensive and non-intrusive.

1.4 Proposed System

The proposed solution integrates physiological and behavioral data to enable accurate mental state detection. It utilizes EEG, EDA, and PPG signals as primary inputs for stress and cognitive load analysis. With the use of facial detection as the second input, there is another behavioral parameter to consider. Patterns of stress and cognitive load can be inferred from the variations in EEG parameters. An adjustable data fusion and machine learning algorithm are utilized in the system. This is done to ensure that the model can analyze different inputs to enhance its predictive accuracy. Various parameters are included in the system to minimize uncertainties in the determination of the mental state of a user. This makes the proposed approach more useful than the single-sensor system [5].

1.5 Project Scope

The project's aim is limited to developing a real-time detection system of mental stress. The system is designed to detect PPG, EDA, and EEG signals. To increase analysis accuracy, facial detection is incorporated as another behavioral input. This project contains data collection, signal preprocessing, feature extraction, and categorization of stress [6]. In addition, several input signals are integrated using data fusion, which is based on machine learning. The system provides feedback to users and analyses stress in real-time. Sensor implementation is restricted to non-invasive and economical sensors that are wearable. Clinical diagnosis and treatment are not part of the system. The system is not designed to substitute professional mental health treatment.

1.6 Assumptions & Constraints

Developers of this project had to rely on some assumptions about environmental and system use. One assumption is that users will wear the EEG device and sensors as instructed. If users place and use the devices as instructed, then data collection will be reliable. It is likely that each user will undergo some sort of baseline calibration. Some of the constraints of the system include the placement, calibration, and servicing of the sensors. Although technology is capable of providing some useful health-related information, it cannot replace a physician's diagnosis. The collection of data may be impacted by external factors that include low lighting, background noise, and excessive movement on the part of the user. The efficiency of the system is likely to be low and is mainly determined by the quality of the signals collected and the active participation of the user. Because of the variable nature of each person, the system may yield different and inaccurate results. It is also assumed that users will follow the recommended guidelines and safety procedures while operating the system [6]. To ensure the system operates at an optimal level and can be relied on over an extended time period, the system is expected to undergo routine servicing and updates.

1.7 Social Benefits

Mental health disorders form a complex sociocultural phenomenon. This project has major positive social consequences by enabling ongoing, discreet, and reasonably priced monitoring of emotional stability. It reduces the risks related to long-term stress and exhaustion by providing preventive insights as opposed to reactive therapies. It enables consumers to take an active role in preserving their psychological health [7].

1.8 Business Plan

This following is the business plan we planned for this specified project in order to present it as a product rather than a project for mental wellness of people.

1.8.1 Business Model Canvas

We provide a highly dependable and actionable mental monitoring solution that competes with clinical tools at consumer pricing by combining inexpensive bio-sensors into a comprehensive AI-driven platform.

1.8.2 Problem

Mental wellness monitoring tools are historically segmented between simple, unreliable consumer apps and extremely expensive, clinical diagnostic setups.

1.8.3 Solution

Clients include students who are under a lot of pressure to perform well academically, remote workers, corporate wellness programs, and ordinary customers who want specific information on their mental habits. makers of smart-watches without explicit EEG sensing and makers of expensive EEG headsets that are too costly for daily use are competitors[8].

1.8.4 Customers

The target customer segments of this project includes the mentally ill people and for technical practitioners who are specialized in mental therapy as they can use our product to fix people in need.

1.8.5 Competitors

Till now there is no similar or competitor worthy solution in the market since the market uses the manual methods or survey of people to identify that the people mental behavior works, there are some applications that works into finding the patterns inside brain but since they also don't involve sensor-based therapy, they're not similar to our project.

1.8.6 Marketing Plan

Our strategy for promoting and marketing the proposed system is followed by the product to be given as the starter to the government professionals to treat some patient from it so that we can get the real-life data and working results and then we can formally start the social media marketing to promote it and direct marketing to promote it to the market expert practitioners[9].

1.8.7 Revenue

When the product will start from the promotion the revenue can be started. The first product design given will be free as the testing product and as the costing in the device was almost 60k we will cost this product on demand installments available to some firms and then we can raise the cost so that now the profit can be aimed.

1.8.8 SWOT (Strength Weakness Opportunities Threats) Analysis

Strengths, combines multiple biological streams enabling strong cross-validation; low- cost hardware strategy. Weaknesses is dependency on precise physical sensor positioning; computational requirements of machine learning integration. Opportunities is expanding field of mental health; extensive linkages with remote monitoring areas and workplace analytics. Threats are the regulations on health parameter capture are changing, and the market is resistant to wearing headgear.

1.8.9 FAB (Features, Attributes, Benefits) Analysis

Features are the real-time ML-powered physiological data fusion across combined inputs. Attributes are the non-invasive, secure, continuous processing, easily interpretable analytics interface. Benefits are the direct mental state understanding driving reduced daily stress and preemptive fatigue recognition.

1.9 Report Layout

This chapter introduced the overarching mental health problem and our multi-sensor resolution. Chapter 2 reviews literature and existing works. Chapter 3 specifies comprehensive system requirements. Chapter 4 explores software design specifications including diagrams. Subsequent chapters reflect upon the implementation, project testing metrics, and project conclusions.

CHAPTER 2: LITERATURE REVIEW/BACKGROUND AND EXISTING WORK

2.1 Background

The project's objective is to develop a multi-sensor system for real-time mental health monitoring. In contemporary society, mental health problems especially stress have grown more common. Conventional monitoring systems frequently depend on just one physiological indicator, such heart rate or EEG. Such single-sensor methods offer little understanding of intricate emotional states. The focus of research has switched during the last ten years to complementary bio-signals for multi-modal monitoring. The accuracy and resilience of stress detection are improved by combining EEG, EDA, PPG, and behavioral data. This is a subset of a global phenomenon related to digital phenotyping and affective computing. Figure **2.1** shows the Full-view architecture of the proposed system.

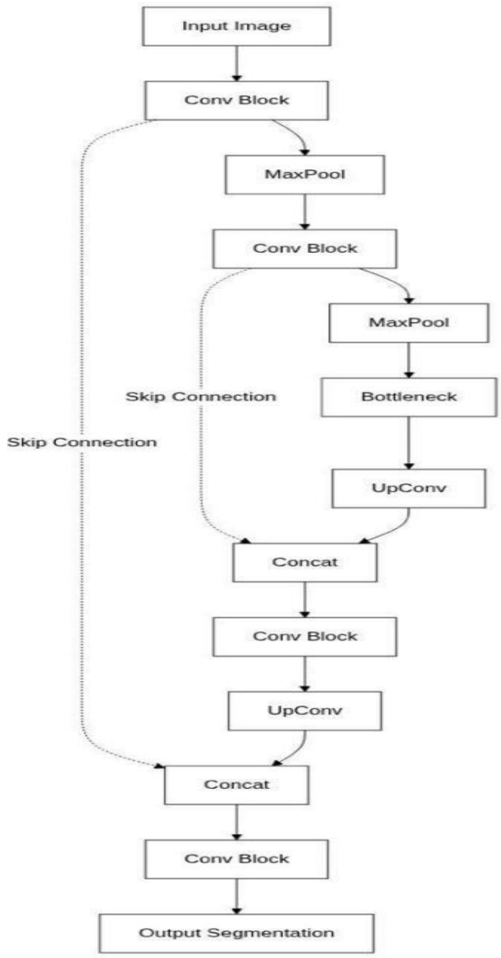

Figure 2.1 Full-view architecture of our system

2.2 Literature Review

This constitutes another development of digital phenotyping and emotional computing. Single-sensor techniques were largely of interest in the earlier stages of research on mental health tracking. An example of one of these methods could be heart rate tracking, which has been clinically shown to map out and identify physiological Stress Response Cycles (especially in the case of the Flight-or-Fight response). Another one of these sensor methods could be EEG, which has been used in research in the past to analyze cognitive load and activity of the brain. Although these techniques have the potential to identify physiological responses to a level of granularity, in the case of heart rate and stress response cycles, these techniques still oversimplify complex emotional phenomena. These techniques are not the most viable for the detection of real time stress in real world scenarios, for the primary reasons of being significantly susceptible to noise and lacking accuracy.

2.2.1 Multi-sensor and Multi-modal monitoring approaches

Recent studies illustrate the importance of integrating multiple physiological signals like PPG, EDA, and EEG. Along with their contributions to stress and emotional state recognition, multi-sensor systems incorporate information from both the central and autonomic nervous systems. Increases in the robustness and accuracy of classification of these methodologies represent their principal advantages. Unfortunately, the cost, size of the hardware, and limited availability for many of the systems hinder their adoption.

2.2.2 Data Fusion and Machine learning techniques

In emotional computing research, data fusion approaches in conjunction with machine learning models have garnered considerable attention. These techniques lower uncertainty and improve prediction reliability by combining information from several bio-signals. Fusion models based on machine learning outperform traditional rule-based systems, according to research.

2.2.3 Non-Invasive and wearable Monitoring Equipment

Wearable technology has revolutionized physiological monitoring by making it possible to collect data continuously without intrusive treatments. The advantages of real-time monitoring, portability, and user comfort are highlighted by research. However, commercial wearables are not made with mental health analysis in mind and frequently rely on single-sensor inputs.

2.3 Literature Summary

From single-sensor systems to multi-sensor, multi-modal approaches, mental health monitoring has clearly evolved, according to the reviewed literature. EEG-only or heart- rate-only devices are examples of single-sensor systems that are simple yet insufficiently accurate and reliable for practical use. Multi-sensor systems that use EEG, EDA, PPG, and behavioral inputs reduce uncertainty and improve prediction reliability using data fusion and machine learning algorithms.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Stakeholders List (Actors)

1. Clear outcomes and objective, continuous self-monitoring immediately benefit end users (individuals, patients).

2. Clinical users: trustworthy, objective data to guide therapy modifications, diagnosis, and patient-specific activities (psychologists, psychiatrists).

3. System Implementers: Code quality, algorithm performance, technical viability, and system stability (Developers, Machine Learning Engineers).

4. Quality Assurance (Supervisors, Researchers): Scientific validation, achieving target accuracy metrics.

5. Organizational Users (Healthcare Providers, Wellness Companies): Scalability of the platform, data security.

3.2 Requirements Elicitation

Requirements were determined through continuous analysis of prior empirical approaches in physiological recording applications.The fundamental benchmarks for successful analysis workflows were deliberately created by confirming limitations on currently used hardware limits and reviewing research on efficient multimodal combinations.

3.2.1 Functional Requirements

FR1 (Data acquisition concurrently): Continuous, synchronized, real-time raw data streams of EEG, EDA, and PPG from the corresponding hardware sensors must be effectively collected by the system.

FR2 (Signal conditioning pipeline): It is necessary to quickly and thoroughly preprocess raw biosignal data in order to eliminate artifacts.

FR3 (Multimodal feature generation and fusion): An intelligent feature fusion operation that combines separate modalities is used by the system.

FR4 (AI Inference): To classify gathered features into stress categories, deep learning and machine learning are used.

FR5 (Visualization): To classify gathered features into stress categories, deep learning and machine learning are used.

3.2.2 Non-Functional Requirements

NFR1: Real-time execution (Performance).

NFR2: High Reliability (Tolerance for sensor noise resulting from limitation on user mobability).

NFR3: Security and Privacy (End-to-End Encryption of biosignal telemetry during storage and presentation).

NFR4: Extensibility (Modular Object-Oriented design that facilitates dropping faulty signal matrices cleanly at runtime).

3.2.3 Requirements Traceability Matric

Each Functional Requirement directly relates to the core issue of cross-verifying measurements to ensure accuracy. Input is captured by FR1, clean input is ensured by FR2, insights that align with user objectives are extracted by FR3 and FR4, and this is then directly communicated to human users by FR5.

3.3 Use Case Design/Description

The principal components consist of a System Model which, by processing data streams, computes parameters indicative of cognitive stressors to present an overall summary in dashboard format; and the User, which displays a set of physiological metrics. Typical applications of this include, but are not limited to, Monitor Status, Review Historical Trends, and Alert Generation on Anxiety Peaks. **Figure 3.3** shows the Use case diagram of the mental health detection system.

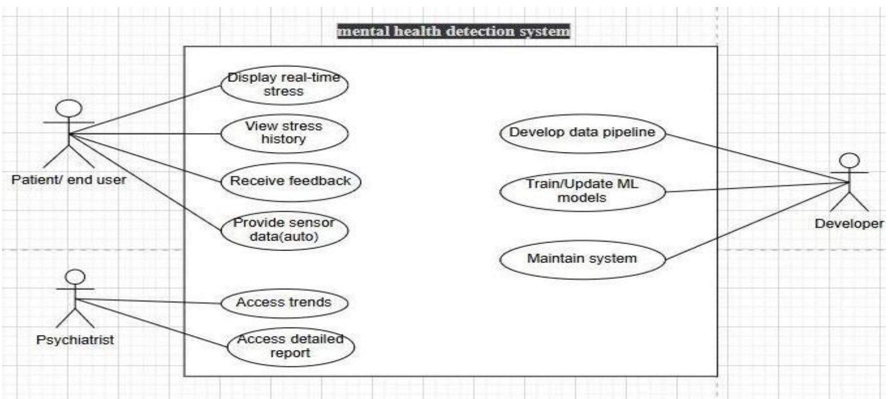

Figure 3.3 Use-case diagram

3.4 Software Development Life Cycle Model

For this project, the Object-Oriented Methodology (OOM) will be employed. Use of this system works well because there are many connected modules, such as database management, machine learning models, sensor data collection, data preprocessing, and user interfaces. Functions such as the addition of multiple sensors can be done using inheritance and polymorphism without the need to redesign the whole system. Meanwhile, OOM provides greater abstraction.

3.5 Specific Requirements (Hardware and Software Requirements)

Following are the hardware and software prerequisites in order to run this proposed solution project.

3.5.1 Hardware Requirement

The hardware requirements necessary for this project are shown in **Table 3.5.1** in the following.

Table 3.5.1 Hardware requirements

Input Devices	Heart Rate Sensor, EDA/GSR Sensor, EEG Headband (Optional but highly recommended), Webcam for optical monitoring
Processor	2.6 GHz or faster
RAM	4 GB minimum (16GB recommended for ML training)
Disk Space	At least 4GB of free SSD storage

3.5.2 Software Requirement

The list of the the required software components for this proposed solution is shown in the **Table 3.5.2** in the following.

Table 3.5.2 Software requirements

Operating System	Windows 10 / Linux
Programming	Python (NumPy, Pandas)
AI Framework	TensorFlow / PyTorch
Backend Integration	Flask / FastAPI
Database Management	MySQL
Computer Vision	OpenCV

CHAPTER 4: SOFTWARE DESIGN SPECIFICATION

4.1 Design Models

There are various design models and methodologies which includes object-oriented design, modular design, etc. We choose object-oriented design for this particular project because of the object (value) needed and classes needed in such a way that we can also work for modular approach.

4.2 Work Breakdown Structure

The workflow is broken down into some compartments including the data acquisition, signal processing, feature extraction, machine learning, database integration and the user interface with the hardware structure of this project.

4.3 System Architecture

The design also entails the frontend visualization component, data processing signals channels, machine learning models and artificial intelligence components in predicting the stresses, as well as the components of interface with hardware sensors and data ingress. **Figure 4.3** shows the System architecture of the proposed mental health detection system.

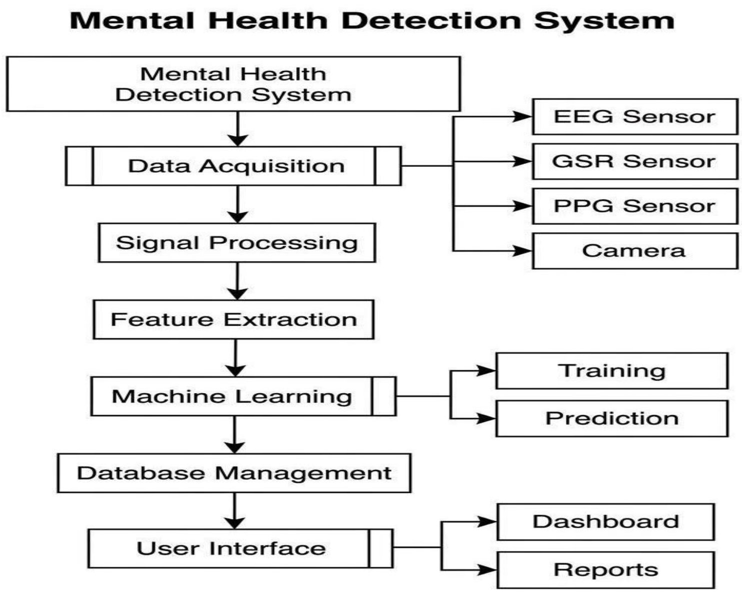

Figure 4.3 System Architecture

4.3.1 Block Diagram

This is the high-level block diagram depicting the major components of our project. **Figure 4.3.1** shows the Block Diagram of the mental health detection system proposed with all procedures starting from data acquisition to predicting mental state

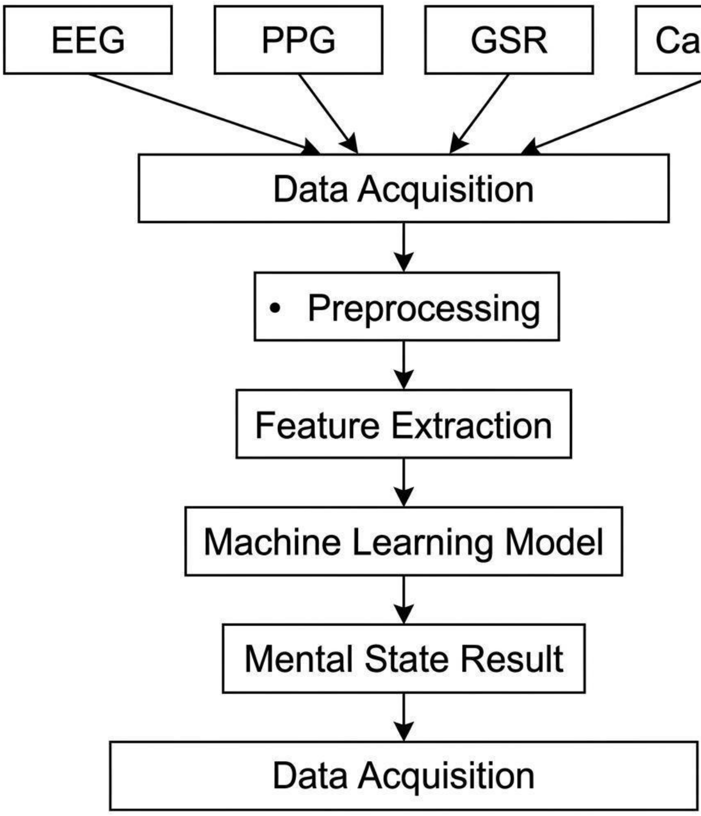

Figure 4.3.1 Block diagram

4.3.2 Component Diagram

This illustrate the organization and relationships of our system components. Figure 4.3.2 shows the Components Diagram of the mental health detection system proposed which shows the inter-relation between different components such as sensor interface, signal processing, feature extraction, machine learning, database and dashboard

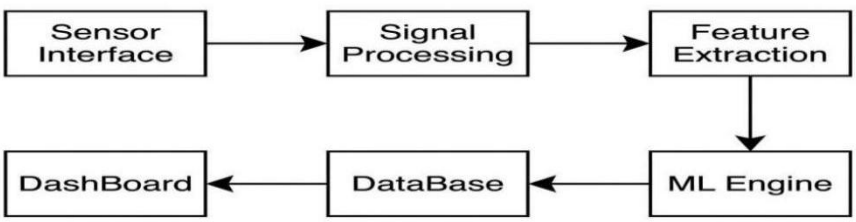

Figure 4.3.2 Component diagram

4.3.3 Software Architecture Diagram

Following **Figure 4.3.3** show the Software Architecture of the mental health detection system proposed with all procedures from Multimodal data acquisition, pre-processing and machine learning based mental state prediction

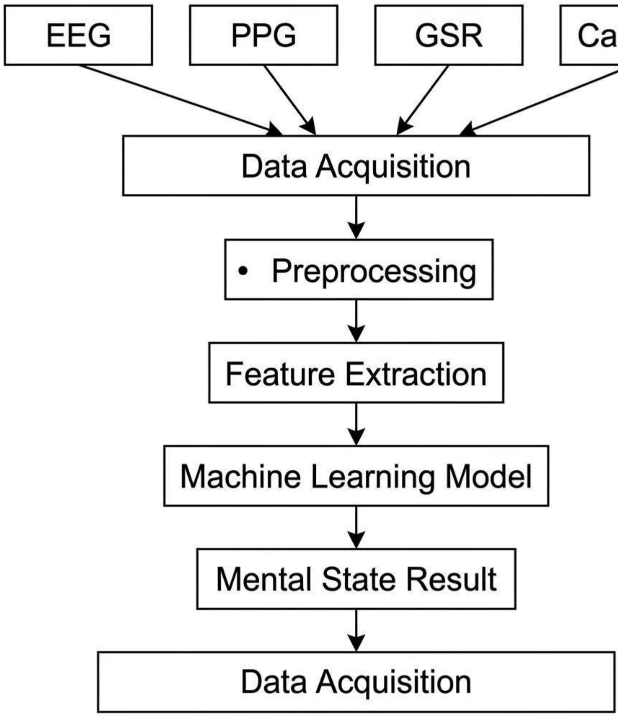

Figure 4.3.3 Software architecture

4.4 Data Representation

This the data representation diagram **Figure 4.4** shows the different layers of the system and how the data was actually processed in the system view.

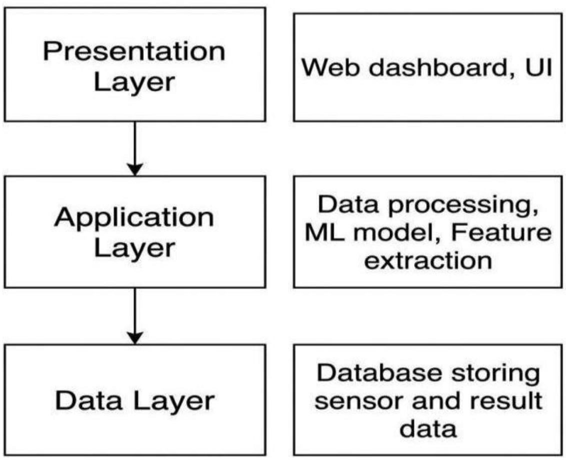

Figure 4.4 Data Representation diagram

4.4.1 Entity-Relationship Diagram (ERD)

This is the ERD according to the need of the database for this required system and **Figure 4.4.1** show the key attributes and its values and show the Entire ERD with its cardinalities:

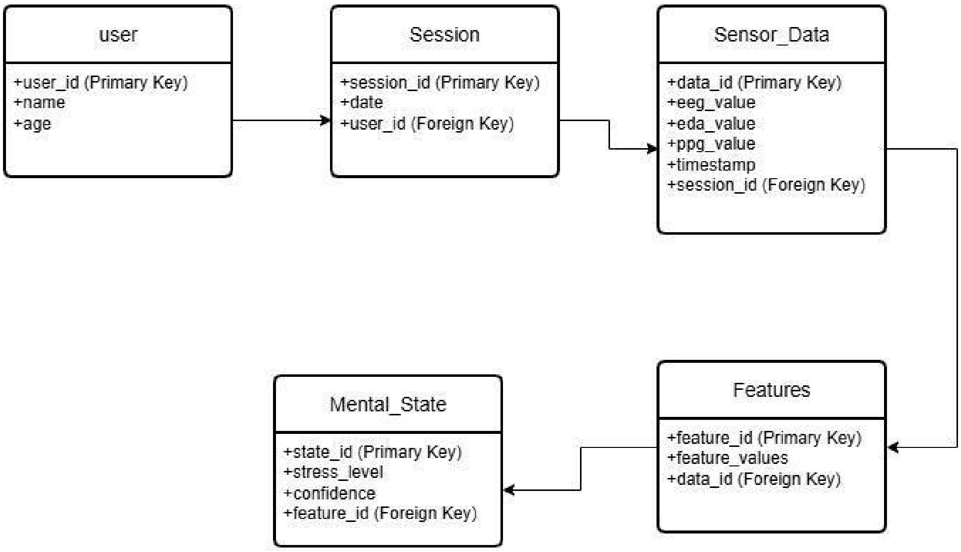

Figure 4.4.1 ERD

4.4.2 UML Class Diagram

This is the UML class diagram in **Figure 4.4.2** depicting the classes, attributes, methods, and relationships of our proposed solution it models the static structure of the software requirements:

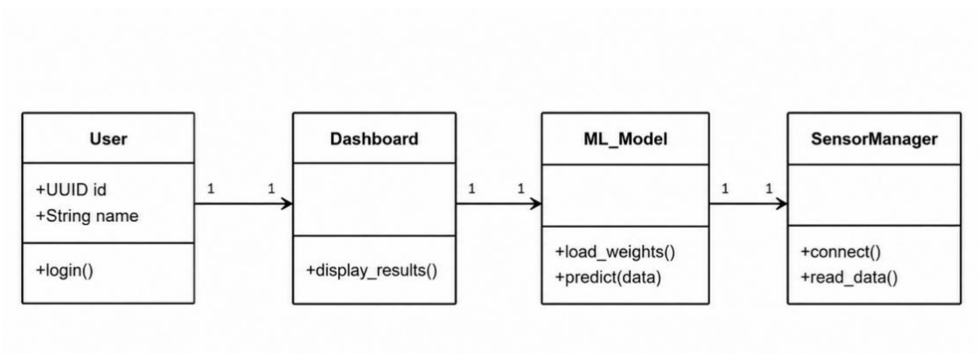

Figure 4.4.2 UML Class diagram

4.4.3 Data Flow Diagram (DFD)

This is the DFD in the **Figure 4.4.3** that illustrate the flow of data through the system processes and show the overall data flow system in the project.

Figure 4.4.3 DFD

4.4.4 Hierarchical Diagram

This is the hierarchical diagram in **Figure 4.4.4** to represent the organization of system components in a tree-like structure this structure explains the design making at any point of logic:

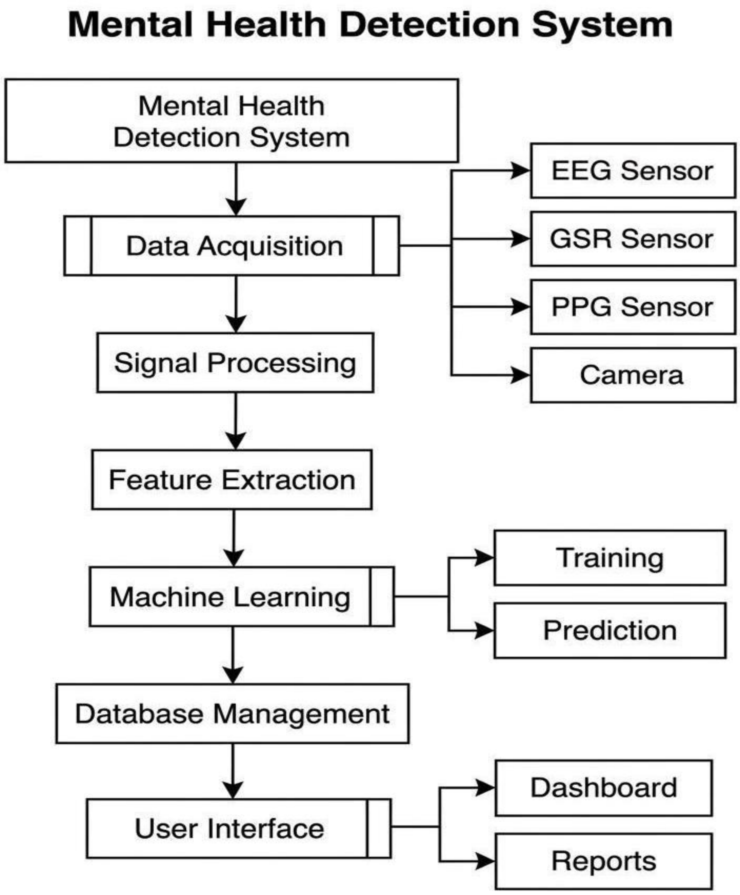

Figure 4.4.4 Hierarchical diagram

4.5 Process Flow/Representation

This is the representation of the flow of the processed of this system:

4.5.1 Flowchart

This flowchart in **Figure 4.5.1** shows the flow of logics as we go from the start to the end of the project system architecture:

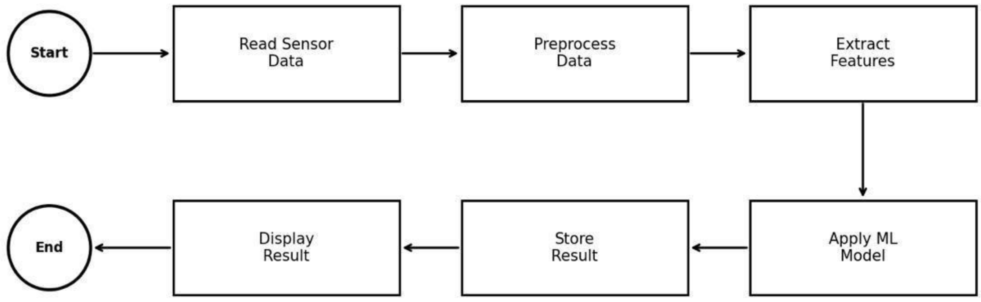

Figure 4.5.1 Process Flow

4.5.2 Sequence Diagram

This sequence diagrams in **Figure 4.5.2** illustrate interactions between objects or components over time. It also show the flow sequence of the project

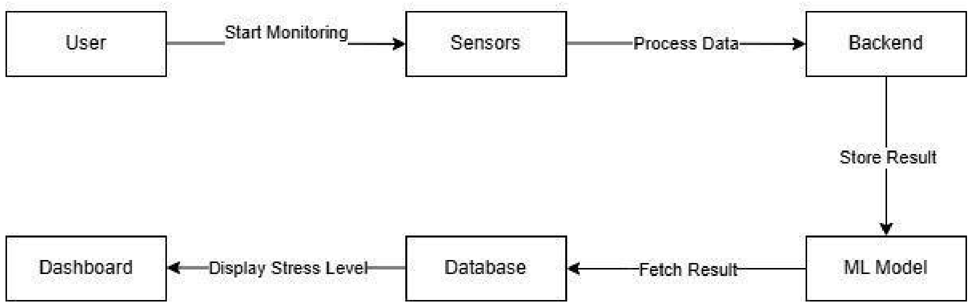

Figure 4.5.2 Sequence Diagram

4.5.3 Activity Diagram

This is the activity diagram in **Figure 4.5.3** that represent the actions, control flows, and concurrent behaviors of this particular project. This show the flow of whole project.

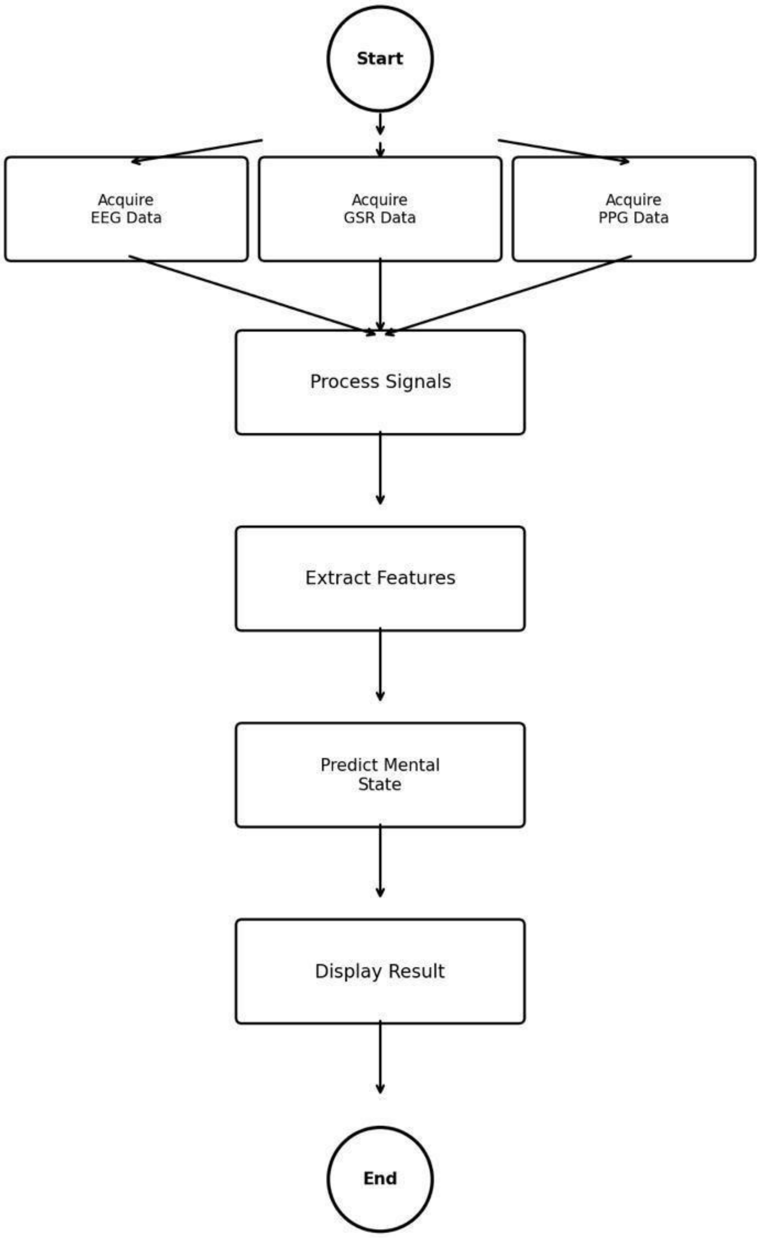

Figure 4.5.3 Activity diagram

CHAPTER 5: IMPLEMENTATION

5.1 Algorithm

Using finite impulse response filtering, the system applies a noise reduction technique to clean the sensors utilized to measure physiological signals. After the restoration of the signal baseline, multi-modal features such as heart rate variability (HRV) calculated from photoplethysmography (PPG) and frequency power bands from electroencephalography (EEG) are statistically extracted. In order to create the present cognitive and emotional status prediction, a machine learning fusion algorithm that is usually based on a Convolutional Neural Network or ensemble classifier receives these parallel feature parameters simultaneously and resolves their interconnections.

5.2 External APIs

In order to maintain latency measurements for real-time monitoring, external dependencies in this iteration primarily rely on local execution runtimes (OpenCV APIs for handling incoming face data vectors) rather than remote cloud connections. The local representational state API that connects hardware measurements to the interface frontend is provided by Flask.

5.3 User Interface

The user interface prioritizes simple insight visualization and offers a minimal dashboard. Stress indicators are reduced to simple measurements in comparison to past baselines. **Figure 5.3** show the whole machanism.

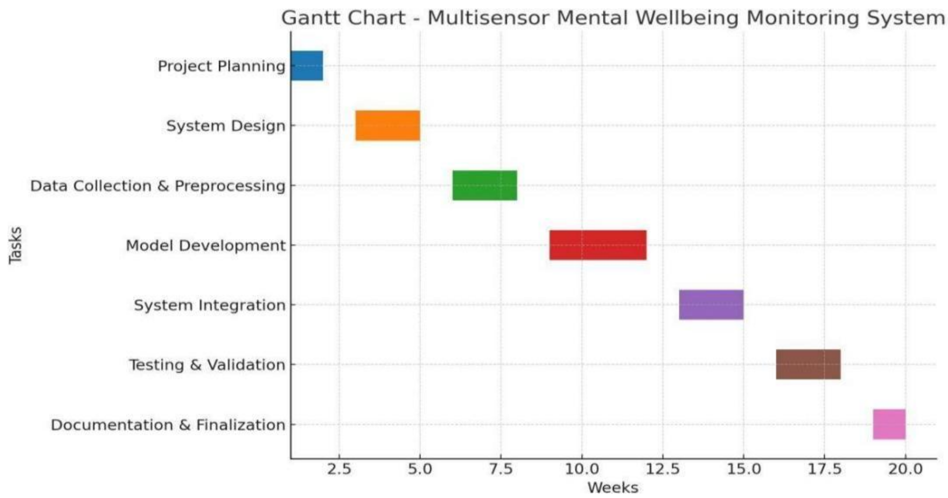

Figure 5.3 User Interface

CHAPTER 6: SYSTEM TESTING

6.1 Manual Testing

Manual testing, which involves the direct application of simulated cognitive burdens to the subjects, is performed to validate the accuracy of the framework. This testing includes measuring and recording the dashboard's delta responses.

6.1.1 System Testing

System testing determines if EDA, PPG, and EEG sensors can be run simultaneously with hardware inputs.

6.1.2 Unit Testing

Unit testing assesses the integrity of the mathematical framework for the noise-reducing temporal filters, ensuring that the intrinsic frequency spectra of the biological input channels are preserved. Integration testing assesses the ability of the system to facilitate flawless interaction among the EEG, EDA, PPG, and facial expression processing components. Performance testing assesses the system's capacity to deal with the parallel processing of multimodal channels with negligible latency. Validation testing determines the stress and emotion classifications against established benchmarks. Security testing evaluates the protection of sensitive physiological data during both the transmission and storage phases. Usability testing assesses the ability of the system's user interface to convey the test results and to facilitate interaction by the researcher and the end user.

6.1.3 Functional Testing

Functional testing verifies the robustness of the classification framework by assessing its ability to parse independent components (e.g. testing the behavior of the OpenFace toolkit in isolation).

6.1.4 Integration Testing

The system was integrated in such a way that the hardware sensor's reading read the data and the user interface read the facial data and the whole system was tested which proceeded with as expected results of a working prototype. 85% accuracy was achieved which produced better results but it can be optimized for the future.

6.1.5 Tool Used

We used pytest for the testing of the model integration and did the manual testing of the fuctionality for the hardware.

6.2 Results & Discussion

Preliminary testing results support the primary hypothesis stating that multiple sensor inputs reduce misclassifications exhibited by single-sensor proxies. Inference consistency is expected to improve by importing larger datasets and by clinical verifications in the future. The proposed framework enhances robustness to greater signal noise and to sensor-specific artifacts. Increased precision in the recognition of emotional states and the assessment of stress in various situations is exhibited by the experiments. The system is expected to be precise, scalable, and implementable with the continued development and real studies. The framework will be cross-validated against various demography. Advanced feature extraction and fusion methods are expected to be game changers in the system's stability and predictive performance. The modular nature of the system will enable the easy incorporation of additional physiological sensors in the future. Extensive evaluation studies will be conducted to further assess system usability, user integration, and the feasibility of real world implementations.

CHAPTER 7: CONCLUSION

7.1 Problems Faced and Lessons Learned

The integration of user movement noise against EDA measurements proved problematic. The TDP detrending method showed us practical applications for digital signal processing and the importance of measuring machine learning bias. Distortions from user movement resulted in baseline drift. Consequently, band-pass filtering and adaptive smoothing were proposed to reduce motion distortions. This example shows the importance of having a fast and reliable preprocessing pipeline in order to deal with the distortions of physiological signals in the field.

7.2 Conclusion

Our wellness monitoring system solves the limitation of existing single sensor systems by integrating several biosensors and computer vision. The system enables the user to assess both the physical and mental aspects of his/her wellness in real time and is also designed to be affordable and easy to use. This system can be implemented in almost any environment and eliminates the need for expensive clinical systems after the initial setup. This system is a flexible and user-friendly way of promoting proactive wellness evaluation in everyday environments.

7.3 Limitations of the Project

Dependence on the Sensor: The placement, calibration, and continual adjustment of the sensors are all critical for the accuracy of the system. Inadequate placement leads to a chain of erroneous predictions. There is a lack of clinical validation, and the technology is not a substitute for a doctor's medical diagnosis. Inconsistency among the users' measurements and reliability of the sensors are affected by the temperature and humidity of the environment and the movement of the user. Due to the variability of the phenomenon, the performance across different users is also inconsistent. Increasing the reliability and robustness of the system, along with the reliable calibration, requires adaptable and corrective methods.

7.4 Future Work

Later versions of the system may increase the diversity of the AI-recognizable psychiatric conditions by expanding the range of recorded biosignals. Further integration of biosensors may improve user comfort, thus allowing longer engagement. Real-world clinical studies are a distant goal for further iterations of the project.

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Pseudo Code

A Code style has been prepared for formatting short excerpts of source code. It is a simple indented, single-spaced style using a fixed font (Courier New) to produce code that appears like the following:

```
1. START MindSense AI System

2. INITIALIZE System Configuration

 IMG_SIZE is set to 48

 SEQUENCE_LENGTH is set to 10

  FORECAST_HORIZON is set to 5

 SENSOR_COLUMNS are defined

 MENTAL_STATE labels are defined

 EMOTION labels are defined

3. INITIALIZE Shared SensorData Object

 GSR

 PPG

 Delta

 Theta

 Low Alpha
```

High Alpha

Low Beta

High Beta

Low Gamma

Mid Gamma

High Gamma

4. INITIALIZE ArduinoReader Thread

Connect to Arduino COM Port

GSR values are READ

PPG values are READ

BPM is calculated

SensorData is updated

5. INITIALIZE EEGReader Thread

Connect to TGAM EEG Device

READ EEG packet

Extract brainwave bands

```
 EEG values are normalized

 SensorData is updated

6. LOAD Trained Machine Learning Models

 Load RNN Sensor Classification Model

 Load Facial Emotion Detection Model

 Load Future Prediction Model

 Load RobustScaler

 Load Label Encoder

7. INITIALIZE Inference Engine

8. INITIALIZE Recommendation Engine

9. CONNECT to Supabase Database

10. INITIALIZE Webcam

 START ArduinoReader Thread

 START EEGReader Thread

 Sensor History Buffer is Created

 Real-Time Monitoring Loop
```

LOOPS

11. READ Current Sensor Values GSR, PPG, Delta, Theta, Low Alpha, High Alpha, Low Beta, High Beta, Low Gamma, Mid Gamma

ADD current reading to Sensor History Buffer

IF buffer size > 10 Then REMOVE the oldest reading

Preprocess the Sensor Data

Convert the readings into feature vector

Scale the values using RobustScaler

Create sequence of 10 timesteps

PREDICT Current Mental State by INPUTting the sequence into RNN Model

PREDICT States: NORMAL, LOW_STRESS, MODERATE_STRESS, HIGH_ANXIETY, PANIC_STATE, DEPRESSION

CAPTURE Webcam Frame

DETECT Face

If a face is detected

The face is cropped.

The face is histogram equalized.

The face is resized to 48 × 48 and the pixel values are normalized.

The face is sent to the Facial Emotion Model.

12. The emotion is predicted to be one of the following:

Angry

Disgust

Fear

Happy

Neutral

Sad

Surprise

The emotion and the confidence level are saved.

If a face is not detected

The emotion is saved as “No Face Detected.”

What follows is the prediction of the future mental state based on the emotion.

13. To predict the future mental state, the latest 10 samples of the emotional state are taken.

14. The samples are sent to the Seq2Seq BiLSTM Model, which predicts the future emotional state for the next 5 time steps.

15. The predicted emotional states are derived based on the following emotional states:

GSR

PPG

Delta

Theta

Low Alpha

High Alpha

Low Beta

High Beta

Low Gamma

Mid Gamma

High Gamma

16. The prediction of the emotional states is used to derive the future mental state.

17. To derive the future mental state, the following are taken into consideration:

If GSR is greater than 30,000 or PPG is greater than 130, the future mental state is defined as PANIC_STATE.

If GSR is greater than 20,000 or PPG is greater than 115, the future mental state is defined as HIGH_ANXIETY.

If GSR is greater than 12,000 or PPG is greater than 100, the future mental state is defined as MODERATE_STRESS.

If GSR is > 6,000 or PPG is > 88, mental state is classified as LOW_STATE_STRESS in the future.

If GSR is < 2,000 and PPG is < 60, mental state is classified as DEPRESSION in the future.

18. In the remaining scenarios, mental state is classified as NORMAL in the future.

19. After mental states are classified, the trend can be evaluated.

The trend is classified as WORSENING when future mental states become worse.

The trend is classified as IMPROVING when future mental states become better.

Otherwise, the trend is classified as STABLE.

20. Using the trend, current mental state, and facial emotion, the following recommendations are offered:

21. If the future mental state is classified as PANIC_STATE or HIGH_ANXIETY, the recommendations listed below are

22. Physical Activity Professional Consultation ELSE Recommend: Maintain Healthy Routine Meditation Regular Sleep END IF ADD Emotion-Specific Suggestions ADD Trend-Specific Suggestions Database Operations

23. STORE Results in Supabase Database SAVE: User ID Timestamp Sensor Values Mental State Confidence Facial Emotion Future Predictions Recommendations Dashboard Update

24. UPDATE Desktop/Web Dashboard DISPLAY: Live GSR Live PPG EEG Bands Mental State Confidence Level Facial Emotion Future Timeline Trend Analysis Recommendations GradCAM Visualization Monitoring Continuation

25. WAIT 1 Second

26. REPEAT LOOP Shutdown Procedure

27. ON Application Close Stop ArduinoReader Stop EEGReader Stop Inference Thread Release Webcam Close Database Connection Clear Buffers

28. TERMINATE All Processes

29. END MindSense AI System